

Quantum Route: Bayesian NFL Betting Engine

Brainstorming & Design Document

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1 Overview

Quantum Route is a Bayesian inference engine for quantifying market inefficiencies in NFL point spreads. The core idea: estimate a posterior distribution over each team’s latent strength using Metropolis–Hastings, propagate that uncertainty through to a predicted game margin, and compare the model’s belief against the market spread to surface bets where the model disagrees with the market *and* is confident about that disagreement. This document is a working brainstorm, not a locked spec — open questions are flagged throughout.

2 Data

Primary source: nflverse’s `games.csv` (game-level, not play-by-play). Relevant fields: final scores, home/away teams, season/week, and the closing `spread_line`. Anything requiring closing-line-value tracking will additionally need the *opening* line, not just closing, if available from the same or a supplementary source — to be confirmed.

3 Model Formulation

3.1 Likelihood options

Three candidates, in increasing order of realism and complexity:

- (a) **Bradley–Terry (win/loss only):**

$$P(\text{home wins}) = \sigma(\theta_h - \theta_a + \eta), \quad \sigma(x) = \frac{1}{1 + e^{-x}}$$

where θ_i is team i ’s latent strength and η is home advantage. Simple, but discards margin information and can’t be compared against a spread market directly — only moneylines.

- (b) **Gaussian score margin:**

$$\text{margin} \sim \mathcal{N}(\theta_h - \theta_a + \eta, \sigma^2)$$

Uses full margin information and predicts a number directly comparable to `spread_line`.

- (c) **Student- t score margin** (fatter tails than Gaussian — more honest about blowout frequency):

$$\text{margin} \sim \theta_h - \theta_a + \eta + \sigma \cdot T_\nu$$

Decision: start with (b) to get end-to-end inference working, upgrade to (c) once the pipeline is validated. Revisit (a) only if moneyline markets become a target.

3.2 Priors

- $\theta_i \sim \mathcal{N}(0, \tau^2)$ for each team — weakly informative, centers all teams at league-average strength before seeing data.
- $\eta \sim \mathcal{N}(\eta_0, \tau_\eta^2)$, with η_0 set from a rough historical home-field-advantage estimate (points).
- $\sigma \sim \text{Half-Cauchy}$ or Half-Normal (must stay positive).
- **Open question:** should priors be pooled hierarchically (division or conference-level shrinkage) rather than fully independent per team? Would help early-season estimates before enough games have accumulated.

3.3 Identifiability constraint

Team strengths are only identified up to an additive constant: $\theta_i = 5, \theta_j = 3$ predicts the same outcomes as $\theta_i = 105, \theta_j = 103$. Without a fix, the chain drifts and never converges to interpretable values. Two options:

- Pin one reference team’s strength to 0.
- Soft constraint via prior: encourage $\sum_i \theta_i \approx 0$.

Decision: required before any MCMC run is trusted — this is the most common silent failure mode for this class of model.

4 Inference: Metropolis–Hastings

4.1 Parameter vector and proposal

Parameter vector: 32 team strengths + η (+ σ if Gaussian/Student- t) $\approx$ 33–34 dimensions.

- **Component-wise** updates (one parameter at a time) — easier to debug, slower to converge.
- **Block** updates (propose all parameters jointly) — faster if tuned well, harder to get right.
- Proposal: random-walk Gaussian, $\theta'_i = \theta_i + \mathcal{N}(0, \delta^2)$. Step size δ is the single most important tuning knob.

Target acceptance rate: roughly 20–50%. Too high $\Rightarrow$ steps too small, chain moves inefficiently. Too low $\Rightarrow$ steps too large, chain rejects almost everything.

4.2 Log-space acceptance ratio

With $\sim 2,700$ games multiplied together, raw likelihoods underflow to zero. Everything is computed in log-space:

$$\log \alpha = \min \left(0, [\log L(\theta' | D) + \log p(\theta')] - [\log L(\theta | D) + \log p(\theta)] \right)$$

Accept θ' with probability $\exp(\log \alpha)$; otherwise retain θ .

4.3 Convergence diagnostics

- Trace plots per parameter (visual check for mixing, no drift/trend).
- Run ≥ 2 chains from different starting points; compare via Gelman–Rubin $\hat{R}$ (target $\hat{R} < 1.01$ – 1.05).
- Discard burn-in; thin if autocorrelation is high.

4.4 Posterior predictive checks

Simulate fake seasons from posterior draws; confirm simulated score distributions resemble real NFL score distributions (mean, variance, tail behavior). This is evidence the model is well-specified, not just that the sampler ran.

5 From Posterior to Market Edge

5.1 Posterior predictive distribution

For an upcoming game, push each posterior draw of $(\theta_h, \theta_a, \eta, \sigma)$ through the likelihood to get a distribution over predicted margin — not a single point estimate.

5.2 Edge signal

Two levels of sophistication:

- **Naive:** compare posterior *mean* predicted margin to `spread_line`.
- **Uncertainty-aware (preferred):**

$$\text{Edge} = P_{\theta \sim \text{posterior}}(\text{predicted margin} > \text{spread_line})$$

i.e., the fraction of posterior mass that agrees the model’s line beats the market’s line. This is the differentiator versus a plain point-estimate regression model.

5.3 De-vigging market odds

American/decimal odds imply a probability that sums to $> 100\%$ across both sides (the vig/overround). Before calling anything a genuine inefficiency, market-implied probabilities must be de-vigged (e.g., proportional or power-method normalization) so the comparison is against the market’s *fair* probability, not its priced probability.

6 Betting Infrastructure

6.1 Paper-trading ledger schema (per bet)

Field	Description
game_id, date, teams	Identifiers
model_predicted_margin	Posterior mean (and interval)
market_spread_line	Line at time of bet
closing_spread_line	Line at market close (for CLV)
edge	Posterior-mass edge signal
odds_taken	Actual price, including vig
stake	Dollar amount (or % of bankroll)
result	Win / loss / push
pnl	Realized profit or loss
bankroll_after	Running bankroll

6.2 Calibration tracking

- **Brier score:** $BS = \frac{1}{N} \sum_{i=1}^N (p_i - o_i)^2$, where p_i is the model’s predicted win probability and $o_i \in \{0, 1\}$ the outcome. Lower is better.
- **Log loss** as an alternative/companion metric.
- **Calibration plot:** bucket predictions by confidence (e.g., 60–70%), check realized win rate in each bucket against the stated probability.

6.3 Closing Line Value (CLV)

The professional-bettor standard metric — lower variance than raw win/loss over a single season, and a stronger signal of real skill.

$$\text{CLV} > 0 \iff p_{\text{bet}} < p_{\text{close}}$$

where p_{bet} is the implied probability of the price taken and p_{close} is the implied probability of the closing price. Consistently beating the closing line is strong evidence of genuine edge, independent of any single game’s outcome.

6.4 Bankroll management and fractional Kelly

Full Kelly fraction for a single bet:

$$f^* = p - \frac{q}{b}, \quad q = 1 - p, \quad b = \text{net decimal odds}$$

Fractional Kelly (required, not optional): bet $f = \lambda f^*$ with $\lambda \in [0.25, 0.5]$. Full Kelly assumes the edge estimate is exact; ours comes from a posterior with real uncertainty, so betting full Kelly on a noisy edge is a known way to blow up a bankroll.

Posterior-aware refinement: rather than plugging in a single point estimate of p , integrate expected log-bankroll growth over the posterior samples themselves:

$$f^* = \arg \max_f \mathbb{E}_{\theta \sim \text{posterior}} \left[\log \left(1 + f \cdot b \cdot \mathbb{I}_{\text{win}}(\theta) - f \cdot \mathbb{I}_{\text{lose}}(\theta) \right) \right]$$

This naturally shrinks stake size when the posterior is wide (model is unsure) even before applying the fractional multiplier λ .

Additional rules to decide in advance, not mid-season:

- Stake as % of *current* bankroll, recomputed after every settled bet.
- Hard stop-loss (e.g., halt betting for the week if drawdown exceeds $X\%$).
- Maximum single-bet stake cap, independent of what Kelly suggests.

6.5 Correlated bet detection

Posterior-implied edges on multiple games are not independent if teams share divisions, opponents, weather, or bye-week patterns. A naive ledger treats every bet as independent and understates true portfolio variance.

Open question: minimum viable version — a simple weekly flag for same-division or common-opponent bets, versus a full covariance-aware bankroll simulation.

6.6 Real-money considerations

- **Legal:** confirm eligibility (age, jurisdiction) directly with a licensed operator before any real-money use — not something to infer secondhand.
- **Tax:** gambling winnings are taxable income regardless of whether a W-2G is issued; log net result per bet, not just an aggregate, to make year-end reconstruction possible.
- **Odds shopping:** if using multiple books, log the actual price taken at each — the same bet at -105 vs. -115 is a materially different long-run edge.

7 Bet Size Optimization Tool

7.1 Objective

Given a recommended game and the model’s posterior, output a recommended stake — not just a binary bet/no-bet signal.

7.2 Core approach

Fractional, posterior-integrated Kelly as defined in Section 6.4, with λ as a tunable global risk parameter.

7.3 Constraints to layer on top

- Max stake per bet (% of bankroll cap, independent of Kelly output).
- Reduce stake for games flagged as correlated with other open bets that week.
- Reject/flag bets below a minimum edge threshold (avoid churning on marginal signals where transaction costs / vig dominate).
- Simulate the recommended stake schedule over historical data (Monte Carlo over posterior draws) to report expected growth rate and probability of a given drawdown — before ever using it live.

8 Validation Roadmap

1. Model + Metropolis–Hastings sampler working end to end, with identifiability fixed and convergence diagnostics passing.
2. Backtest on held-out seasons (train on season $N - 1$, predict season N) — report calibration, Brier score, and simulated CLV.
3. Paper-trade forward in real time for a full slate of games (no money moving) — confirm calibration and CLV hold up out-of-sample, not just in backtest.
4. Define explicit, written go/no-go criteria *before* moving to real money (e.g., minimum sample size, positive CLV over N weeks, drawdown within tolerance).
5. Real money, starting at minimal stakes, with the bet-size optimizer active from day one.

9 Open Questions / Decisions Checklist

- ☐ Gaussian vs. Student- t margin likelihood — decide after first working Gaussian version.
- ☐ Hierarchical (pooled) priors on team strength, or fully independent?
- ☐ Component-wise vs. block Metropolis-Hastings updates.
- ☐ Reference-team pinning vs. soft-sum-zero constraint for identifiability.
- ☐ Minimum viable correlated-bet handling.
- ☐ Data source for opening lines (needed for full CLV tracking, beyond closing line alone).
- ☐ Global risk parameter λ for fractional Kelly — fixed, or itself tuned via backtest.
- ☐ Static plots/README vs. a lightweight interactive dashboard — revisit only after the core model and backtest are solid.

10 Resume / Interview Framing Notes

- Hand-rolled Metropolis–Hastings (vs. PyMC/NumPyro) is higher-signal precisely because it proves understanding of the acceptance ratio, not just an API call. Mention faster alternatives (NUTS/HMC) only as “aware of, chose not to use for X reason.”
- Calibration and CLV are more sophisticated metrics than raw win rate — lead with these in any writeup.
- The core differentiator to emphasize: uncertainty-aware position sizing derived directly from posterior samples, not a point-estimate model with a bet size bolted on afterward.